# Multi Dimensional Array(2-D Array)

## Programs

1. **Write a program in 'C' to print 2-D Array.**

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=0,j=0;
int A[4][3];
clrscr();
printf("enter 2-D array elements");
for(i=0;i<4;i++)
{
For(j=0;j<3;j++)
{
scanf("%d"&A[i][j]);
}
}
Printf("Array elements are");
for(i=0;i<4;i++)
{
 for(j=0;j<3;j++)
{
   printf("arr[%d] [%d] = %d \n",i,j,arr[i][j]);
 }
}
getch();
}
```

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2. **C Program to read a matrix and find sum, product of all elements of two dimensional (matrix) array**

**Solution**

```
#include <stdio.h>

#include<conio.h>




void main()

{

        int matrix[10][10];

        int i,j,r,c;

        int sum,product;


        printf("Enter number of Rows :");

        scanf("%d",&r);

        printf("Enter number of Cols :");

        scanf("%d",&c);


        printf("\nEnter matrix elements :\n");

        for(i=0;i< r;i++)

        {

                for(j=0;j< c;j++)
```

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```
        {
                scanf("%d",&matrix[i][j]);
        }
    }



    sum         =0;
    product     =1;
    for(i=0;i< r;i++)
    {
        for(j=0;j< c;j++)
        {
            Sum=sum + matrix[i][j];
            Product = product * matrix[i][j];
}


    }

    printf("\nSUM of all elements : %d \nProduct of all elements :%d",sum,product);
    getch();
}
```

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**Programs for practice.**

1. **Write a program in ‘C’ for addition of matrix.**
2. **Write a program in ‘C’ for subtraction of matrix.**
3. **Write a program in ‘C’ for multiplication of matrix.**
4. **Write a program in ‘C’ for transpose of matrix.**

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